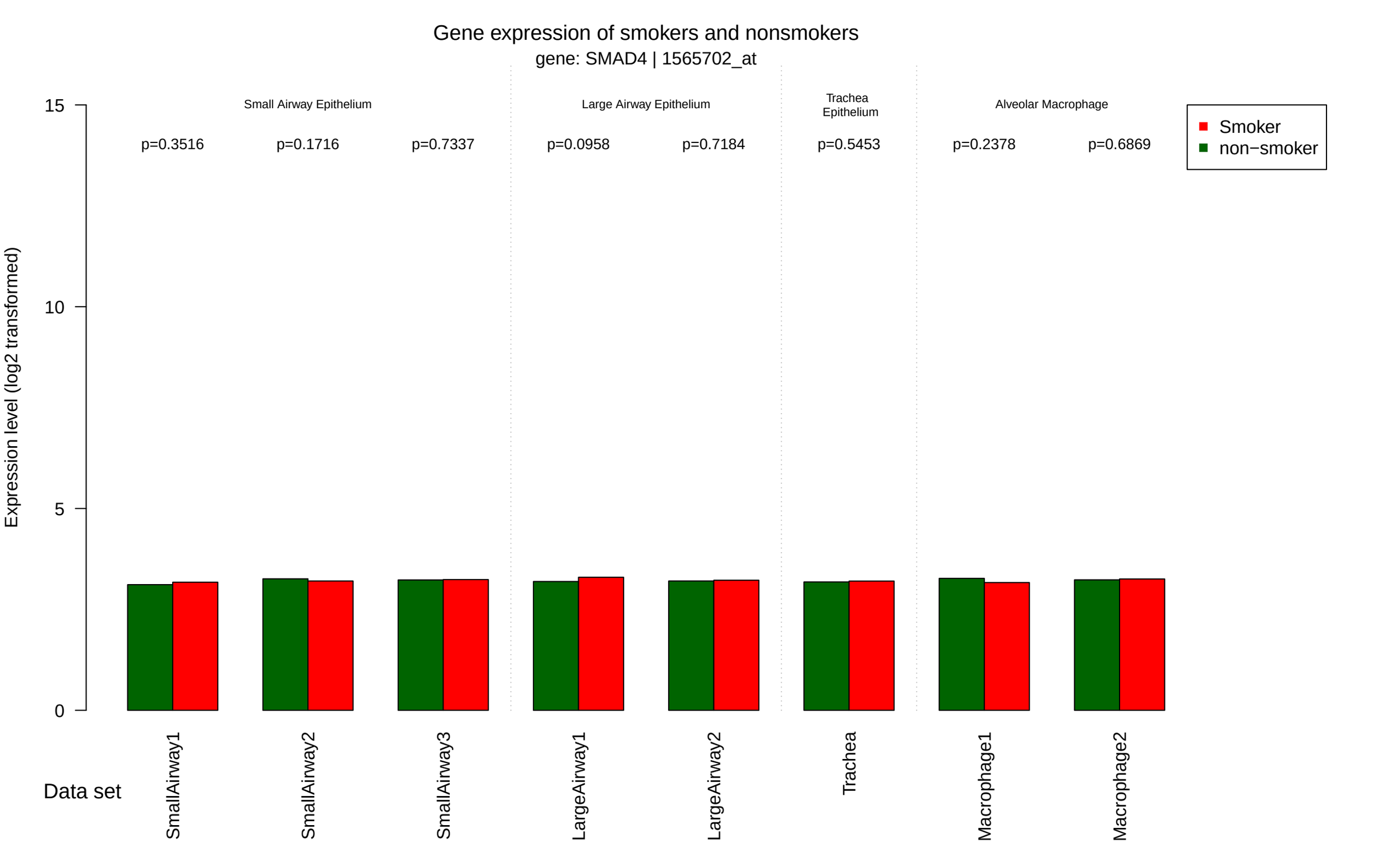
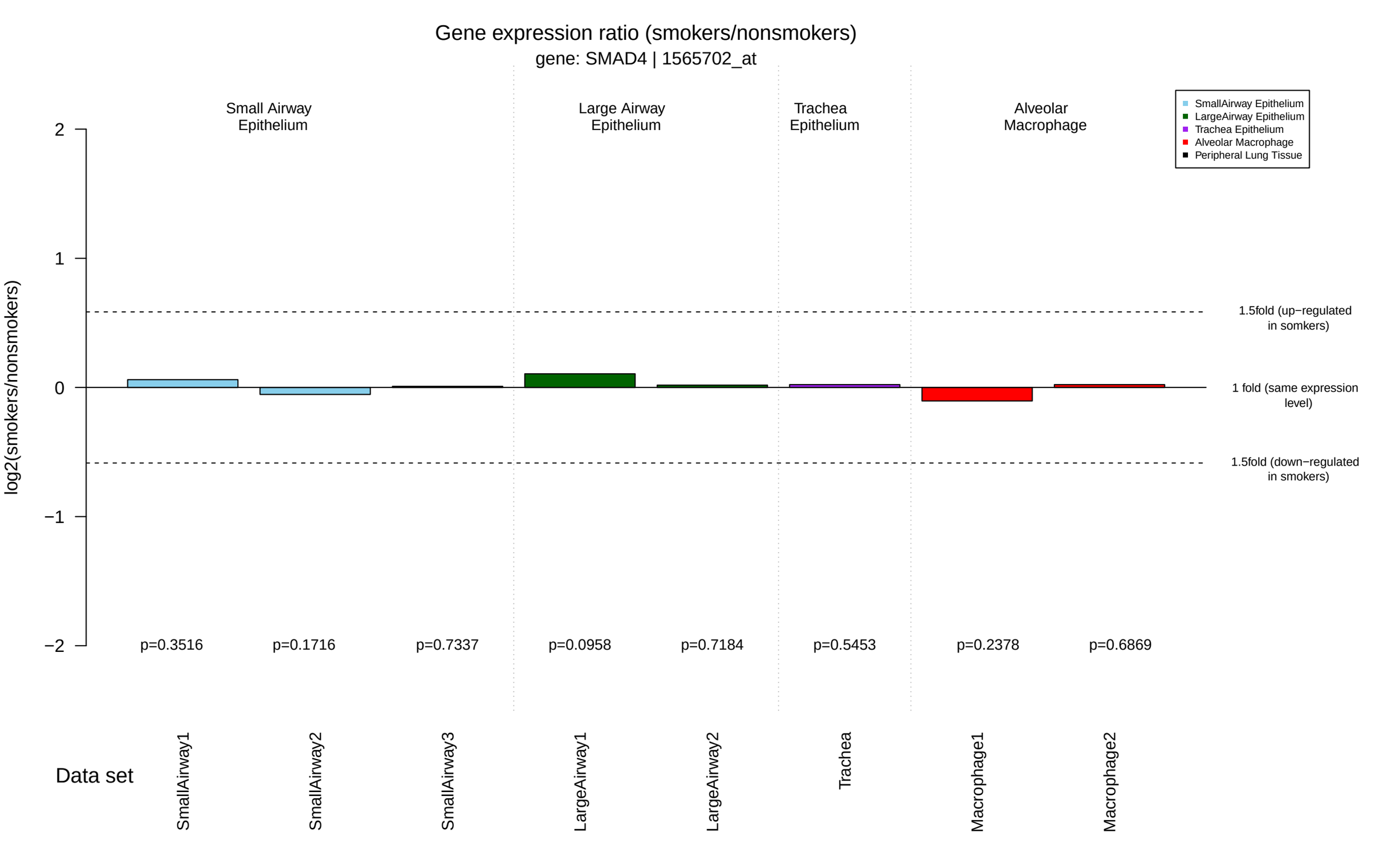
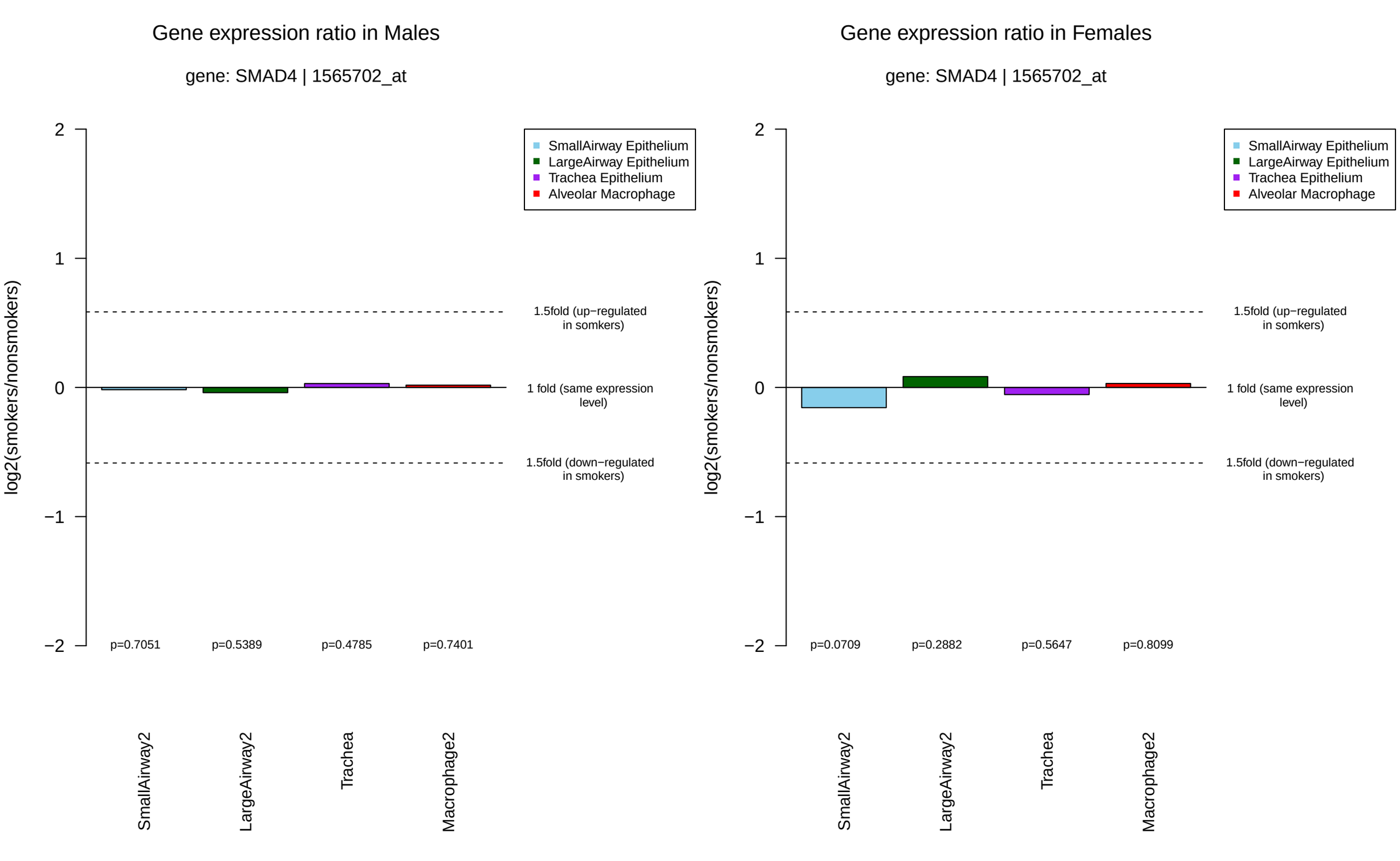
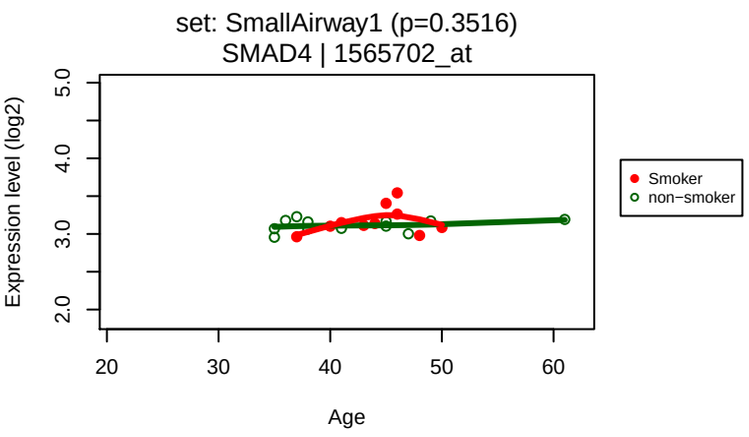
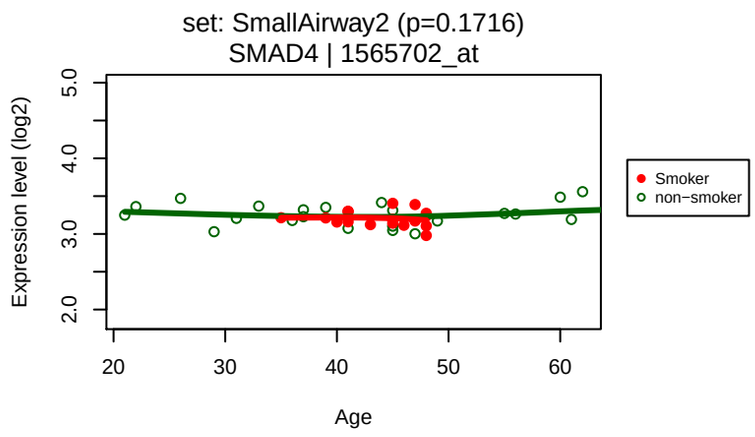
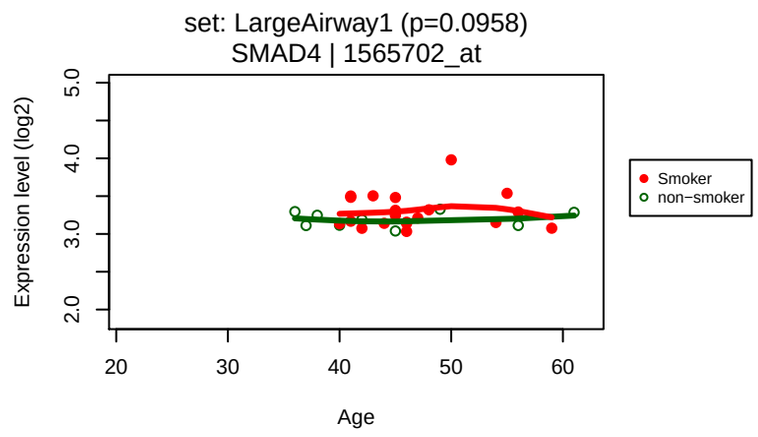

Gene expression of smokers and nonsmokers
gene: SMAD4 | 1565702_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3516
p=0.1716
p=0.7337
p=0.0958
p=0.7184
p=0.5453
p=0.2378
p=0.6869
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

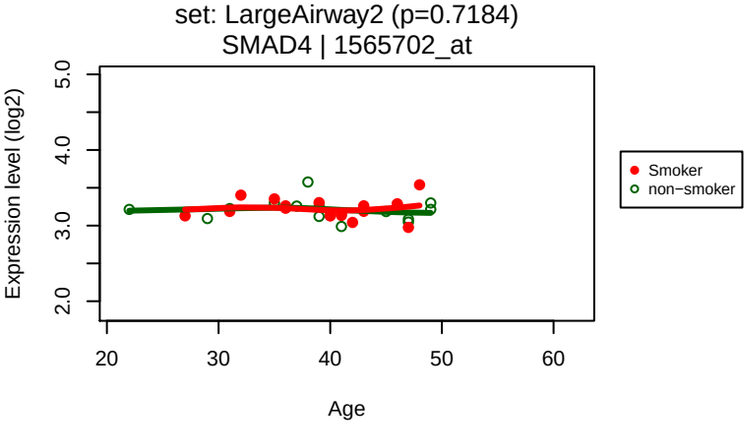

Gene expression ratio (smokers/nonsmokers)
gene: SMAD4 | 1565702_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3516
p=0.1716
p=0.7337
p=0.0958
p=0.7184
p=0.5453
p=0.2378
p=0.6869
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

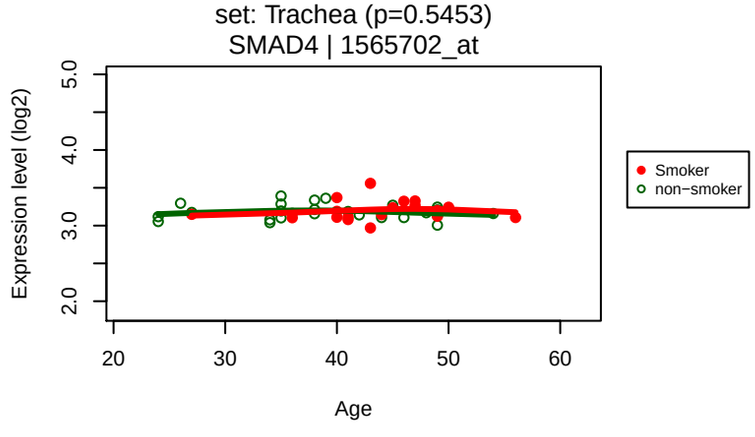

Gene expression ratio in Males
gene: SMAD4 | 1565702_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7051
p=0.5389
p=0.4785
p=0.7401
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD4 | 1565702_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0709
p=0.2882
p=0.5647
p=0.8099
SmallAirway2
LargeAirway2
Trachea
Macrophage2

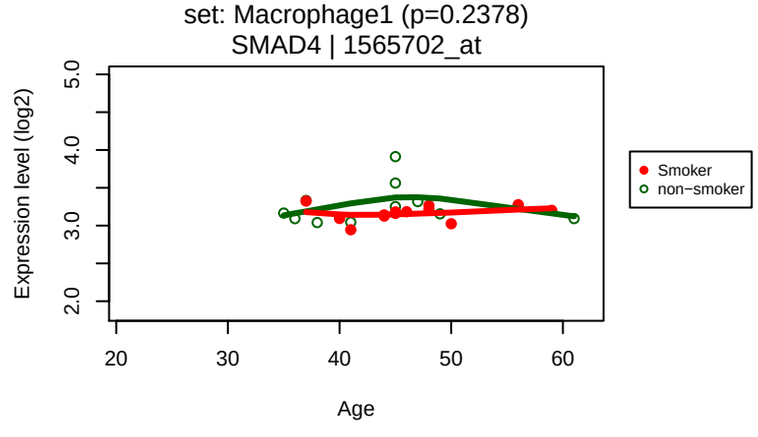

set: SmallAirway1 (p=0.3516)
SMAD4 | 1565702_at
Expression level (log2)
Age
Smoker
non−smoker

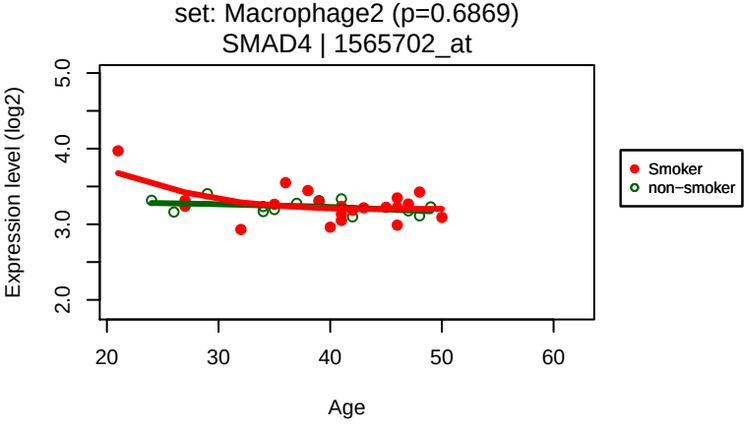

set: SmallAirway2 (p=0.1716)
SMAD4 | 1565702_at
Expression level (log2)
Age
Smoker
non−smoker

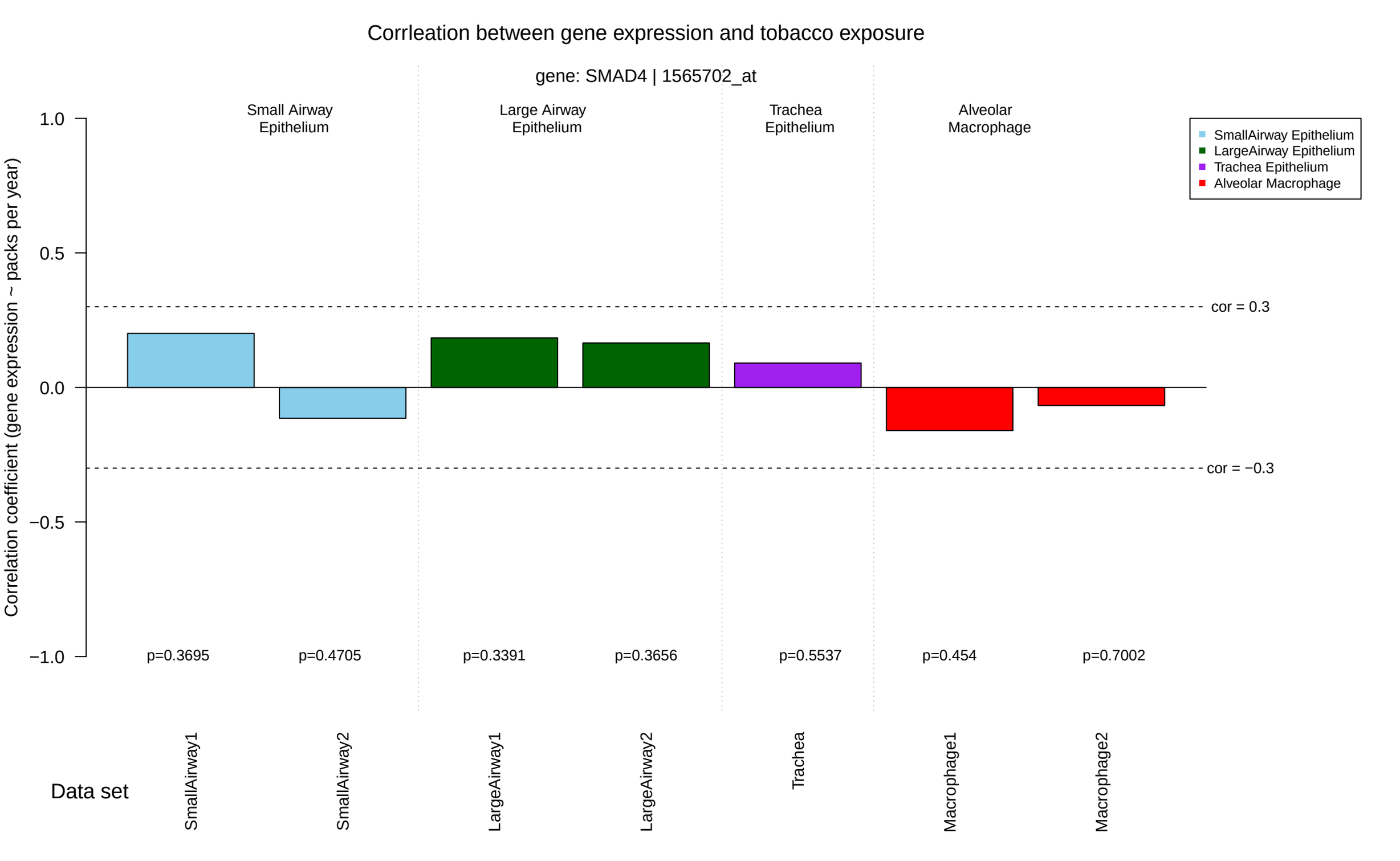

set: LargeAirway1 (p=0.0958)
SMAD4 | 1565702_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7184)
SMAD4 | 1565702_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5453)
SMAD4 | 1565702_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2378)
SMAD4 | 1565702_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6869)
SMAD4 | 1565702_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD4 | 1565702_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3695
p=0.4705
p=0.3391
p=0.3656
p=0.5537
p=0.454
p=0.7002
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2